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Supplementary Material: Agentic Autoresearch for CT Reconstruction

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ABSTRACT

This document contains the supplementary material for *Agentic Autoresearch for CT Reconstruction*. It provides the full solver taxonomy and per-method configuration (S1), the Mayo-LDCT data, split, and geometry pipeline (S2), the breast split and train/test leakage audit (S3), the Mayo test board and significance (S4), the parameter-efficient recombination study (S5), the noise model and no-retrain protocol (S6; the full boards and the framework-axis reading are in the main text), an extended account of agent behavior (S7), and additional results with data and code availability (S8).

S1 | Solver taxonomy and per-method configuration (Table 1, full)

We place the 26 methods on two axes (Table S1) so that the reversal in the main text can be read off a method's *position* rather than its name. Axis A is *physics engagement*: how much the forward operator $\mathbf{A}$ is used at inference. It runs from *none* — a purely image-domain map that never revisits the sinogram — through a *single* data-consistency step, to an *in-loop* unrolled scheme that re-applies $\mathbf{A}$ at every iteration, to a full *per-scene* fit that optimises one network per scan. Axis B is the *prior source* $\mathcal{R}$: where the regularising knowledge comes from. *Hand-crafted* priors (total variation, bilateral filtering) encode smoothness analytically and have no trainable content to overfit. *Supervised* priors are trained on paired low-dose/full-dose examples. *Self-supervised* priors (Noise2Inverse) learn without a clean reference by splitting the measured projections into complementary subsets. *Generative* priors are diffusion or rectified-flow models that synthesise plausible detail. *Implicit/per-scene* priors (neural fields, Gaussian splatting) fit a coordinate network to each individual scan rather than storing a pixel grid. A *foundation* prior is a large network pre-trained once on generic data and applied zero-shot. Each row of Table S1 names a family, its representative solvers, the data-consistency operator $\mathbf{D}$ each uses, the prior $\mathcal{R}$, the two axis positions, and a source citation; the final row is our own recombination (Section S5), deliberately a hybrid — filtered data-consistency, a learned primal-dual combination, and a hand-crafted bilateral post-filter.

Full per-solver `CONFIG` defaults, parameter counts, and design notes are in each solver's design document under `pentathlon/demo_dl_reference/`, and the complete per-iteration observation records — configuration, reconstruction, and metric for every run — are under `docs/runs/`. Because every solver is grounded in the same differentiable projector, two methods in the same Axis-A/Axis-B cell share the same relationship to the measurements and — as the reversal shows — the same robustness behaviour under noise. A method's cell therefore predicts, before any noise is added, whether it will be brittle or robust.

S2 | Mayo-LDCT — data, split, geometry pipeline, and solver implementation

Data and geometry. We use real helical low-dose CT from the 2016 AAPM Low-Dose CT Grand Challenge.[1] The raw acquisition is helical, but our differentiable projector and every solver operate in 2-D fan-beam geometry, so the helical data cannot be used directly. We therefore rebinned helical to fan-beam by single-slice rebinning,[2] which resamples the helical cone-beam rays onto a stack of transaxial fan-beam slices. Getting this right — the detector geometry, the redundancy weighting, and the slice interpolation, so that the low-dose FBP baseline is physically correct rather than merely plausible — was the single hardest engineering task in the study. The differentiable fan-beam projector used at every training and inference step is PYRO-NN,[3] which supplies a matched forward/back-projector pair with analytic gradients, so that the same operator **A** appears in the loss, in every unrolled solver, and in the hr metric.

Effort accounting. The entire repository is agent-generated — the geometry pipeline, all solvers, and the benchmark harness were written by the agent; the human instructed rather than hand-coded. Two cautions therefore apply when reading effort from the version-control history: most commits are the agent's own automated output, and even a source-changing commit is often an autonomous edit made during a search rather than a human touchpoint. Of the 1,785 commits, 1,214 (68 %) change only results or dashboard data with no source edit — automated provenance, no human interaction — while 571 touch source code.

The geometry pipeline needed the most steering. Making the helical-to-fan rebinning and its calibration physically correct (`helix2fan`, the rebinning scripts, the FBP-baseline validation) was the hardest problem: 40 source-code commits over 14 active days, each backed by many rebinning-and-validation runs, so it dominated the human supervision despite being a small share of the commits.

Solvers and benchmarking needed little. The agent onboarded 13 of the 26 established solvers across two days (15–16 May 2026) and the rest in small batches (Table S2); the three-week benchmark campaign is dominated by automated per-iteration provenance (roughly two-thirds of its commits change only results), and the agent ran ~1,100 iterations for ~134 GPU-hours, largely in parallel.

A rough human gauge. Costing each source-changing commit at an estimated 7.5 minutes — a rate from comparable projects — the 571 code commits give an order-of-magnitude *upper* bound of ~70 hours of human steering across the whole two-month project; it is an upper bound because the agent authored many of those commits autonomously. The calendar span itself overstates the effort: it was stretched by the supervisor's travel and by intermittent, part-time attention, not continuous work.

Split. We use a fixed patient-level split so that no patient appears in more than one partition: train L145/186/209/219, validation L277, test L014/056/058/075/123. Splitting by patient rather than by slice matters, because adjacent slices of one patient are highly correlated; a slice-level split would leak near-duplicate anatomy between train and test. Test scores are the per-patient mean ± standard deviation over the five test patients ($n = 5$), and this small n is the direct cause of the low statistical power discussed in Section S4. Geometry details and rebinning validation are in the repository. The Mayo challenge is one of a family of AAPM reconstruction grand challenges spanning sparse-view, spectral, truth-based, and metal-artifact tasks.[28,29,30,31,32]

Table S2 gives, per established solver, the first commit touching its reference file (repository-local time zone) and the number of commits that touched it. The remainder onboarded after 16 May came in small batches as the study grew: diffusion-reconstruction (×2, 17 May), RAM (19 May), Wu-2015-trainable and DD-bilateral-supervised (22 May), learned-primal-dual and TV-iterative-supervised (23 May), the four fast-diffusion variants (20 June), and manduca-bilateral (23 June). Several solvers share a reference file and were introduced together, so per-file counts are approximate; the clear signal is that most methods went from paper to running code within a day.

S3 | Breast split and the train/test leakage audit

Split. The Breast CT Challenge releases a public pool of 4000 training cases; its official validation and test truths are withheld for scoring, so we cannot use the challenge's own held-out sets. We therefore re-partitioned the public pool ourselves into train (3600), validation (200), and test (200). The partition is deterministic — a fixed seed (20260703) assigns each case exactly once — and the three subsets are disjoint. Because the truths in our test split are known to us, the headroom metric can be evaluated directly on it; the price is that we must then prove no test information reaches training. We did so two independent ways.

Leakage audit.

1. *By construction.* The public 4000-case pool is partitioned by case index under a fixed seed (20260703): the train set is a fixed block and the validation and test sets are carved from the disjoint remainder, each case assigned exactly once and recorded in the split manifest (`scripts/restage_breast_valtest_split.py` verifies index-disjointness). No index can appear in more than one subset, so the splits cannot straddle.
2. *By mechanism.* The training data loader is wired to read only the `train` split; the redirect to the test split fires exclusively inside the evaluation path. No code path exposes a test image during optimisation, so training cannot see test even in principle.

Selection during the search used only the validation split; the reported numbers are a single evaluation of the selected iteration on the held-out test set, so the leaderboard is not tuned on test. The high absolute hr reflects an easy, well-posed, incompleteness-limited problem, not memorisation.

Why breast hr is high, and not comparable to Mayo. The Breast CT Challenge data is noiseless by design; its RMSE floor is zero. Breast is incompleteness-limited (128 of ~1000 views). Mayo is noise-limited (real quantum noise). The two absolute hr scales are therefore not comparable; Table S3 contrasts the two problems side by side.

S4 | Mayo test board and significance

The board. The champion is ITNet v1 at test hr 0.376, averaged over the five test patients. We test differences with a paired comparison on the per-patient hr values: for each pair of solvers we form the five per-patient hr differences and run a paired t-test, with a Wilcoxon signed-rank test as a distribution-free cross-check. On this basis the top of the board is a statistical tie — ITNet v1, ITNet v2, and U-Swin are indistinguishable at the 5 % level, and the compact param-efficient solver joins this group at the 1 % level ($p = 0.02$–0.027).

Why the top tier ties at $n = 5$. With only five test patients the paired t-test we use has little power, so a difference clears significance only if it is large *and* consistent across patients. It

TABLE S1 | Full solver taxonomy. Axis A = physics engagement; Axis B = prior source. The final column cites the source(s) for each family. Abbreviations: DC, data consistency; N2I, Noise2Inverse; rectified-flow prior, a fast diffusion-type generative prior; FoE, field of experts (a learned filter-bank image prior).

Family	Representative solver(s)	Data-consistency D	Prior $\mathcal{R}$	Axis A	Axis B	Ref.
Classical iterative	tv-iterative, manhart-PWLS-TV	filtered / SART grad	total variation	in-loop	hand-crafted	[4,5]
Image-domain denoiser	manduca-bilateral, DD-UNet sup.	none (post-FBP)	bilateral / U-Net	none	hand-cr. / sup.	[6,7]
Unrolled learned-iterative	ITNet v1/v2/v3, U-Swin	in-loop DC	learned CNN / transformer	in-loop	supervised	[8,9,10,11,12,13]
Variational network	Hammernik-2017, Hammernik-VN	in-loop DC	field-of-experts	in-loop	supervised	[14,15]
Learned primal-dual	learned-primal-dual (LPD)	filtered + learned dual	learned combination	in-loop	supervised	[16]
Dual-domain	DD-supervised, DD-bilateral, DD-N2I	image + sinogram	sup. / bilateral / self-sup.	single/in-loop	sup. / self-sup.	[17,18]
Generative prior	fastdiff flow/wavelet ×4	DC + diffusion	rectified-flow prior	in-loop	generative	[19,20,21,22]
Per-scene implicit	NAF, R^2-Gaussian	per-scene fit	implicit representation	per-scene	implicit	[23,24,25]
Foundation zero-shot	RAM	single DC	frozen foundation model	single	foundation	[26]
Band decomposition	Wu-2015, Wu-2015-trainable	band-weighted DC	frequency-band residual	in-loop	hand-cr. / sup.	[27]
Recombination (ours)	**param-efficient**	**filtered + LPD combine**	**bilateral post-filter**	**in-loop**	**hybrid**	—

TABLE S2 | Implementation record for the 26 established solvers (every benchmarked method except the param-efficient recombination), from the git history of the reference implementations. "First" is the first commit touching the solver's file (repository-local time zone); "#c" is the number of commits that touched it. The four fast-diffusion variants share two files and the two diffusion-reconstruction variants share one file (grouped).

solver	first	#c	solver	first	#c
itnet	May 15	14	manduca-bilateral	Jun 23	1
itnet-v2	May 15	13	wu-2015	May 16	10
itnet-v3	May 16	11	wu-2015-trainable	May 22	6
uswin	May 16	10	dd-n2i	May 15	14
dd-supervised	May 15	14	dd-bilateral-sup	May 22	4
learned-primal-dual	May 23	3	dd-bilateral-n2i	May 15	16
hammernik-2017	May 16	10	ram	May 19	9
hammernik-vn	May 16	11	naf	May 16	12
tv-iterative	May 15	10	r2gaussian	May 16	12
tv-iterative-sup	May 23	4	diff-recon (×2)	May 17	14
manhart-pwls-tv	May 15	12	fastdiff (×4)	Jun 20	2

TABLE S3 | Breast versus Mayo: the two problems are limited by different things, so their absolute hr scales are not comparable.

	Breast CT	Mayo
data	noiseless, ideal	real low-dose, quantum noise
bottleneck	data incompleteness	noise / dose
RMSE floor	0	> 0
best hr	∼0.89	∼0.38
dominant lever	data-consistency / known operator	denoising

does separate such gaps at $n = 5$: versus the champion, dual-domain-supervised reaches $p = 0.0001$, ITNet v3 $p < 0.0001$, Hammernik-VN $p = 0.003$, and even the compact solver $p = 0.02$. The differences *within* the top tier, by contrast, are both small and patient-to-patient noisy, so ITNet v1/v2/U-Swin stay above 0.05 (and, with the compact solver, above 0.01). Significance here tracks consistency, not mean rank: dual-domain-supervised sits a *smaller* mean gap below the champion than the compact solver, yet is far more significant, because its per-patient deficit is near-constant while the compact solver's is noisy. A distribution-free Wilcoxon signed-rank cross-check is blunter still at $n = 5$: its smallest attainable two-sided p is $2/2^5 = 0.0625$, so two-sided it cannot reject for *any* pair (one-sided floor $1/2^5 = 0.03125$). Hence "a tier, not a winner" is not a hedge

TABLE S4 | Paired t-test p-values versus the champion (ITNet v1), $n = 5$ patients, one test per metric; **bold** marks $p < 0.05$. No multiple-comparison correction is applied. On hr, ITNet v2 and U-Swin tie with the champion at the 5 % level and param-efficient joins the tie at the 1 % level.

vs ITNet v1	hr	SSIM	PSNR	RMSE
ITNet v2	0.264	**0.042**	0.279	0.231
U-Swin	0.738	**0.006**	0.591	0.941
DD-supervised	**0.0001**	**0.004**	**0.0004**	**<0.001**
param-efficient	**0.022**	**0.027**	**0.020**	**0.025**
ITNet v3	**<0.001**	**0.004**	**0.0001**	**0.002**
Hammernik-VN	**0.003**	**0.003**	**0.005**	**0.001**

but the limit of what five patients resolve *for gaps this small* — which is why we report the per-case mean ± standard deviation rather than a bare ranking. The pairwise p-values are in Table S4 and the corresponding forest plot in Figure S1; per-patient raw values and the full solver×metric significance matrix are in the repository (`docs/runs/mayo_significance_stats.md`). The full board (all 26 solvers, hr/SSIM/RMSE) is in the main text.

S5 | The parameter-efficient recombination study (search arc)

The agent recombined the method parts into a minimal-parameter solver on each dataset. Each climb was made of genuine six-box iterations, all under the 20-min budget, and the optimal recombination differs by problem. We present the Mayo arc first, then breast.

Mayo (low-dose, denoising). The seed is a learnable CNN regulariser in a tied prox–DC unroll. Switching the regulariser *family* from a CNN to a field-of-experts (FoE) filter bank set the agent-alone ceiling at hr ≈ 0.25 (Table S5). The break came from the projection domain: prepending a noise-adaptive Manduca bilateral filter to the sinogram, then extending it to a multi-scale bank, lifted val hr to 0.40. Halving the FoE bank yields a 969-parameter Pareto pick (test hr 0.324) — the reported Mayo compact solver.

Breast (128-view sparse). Key steps, all under the 20-min budget:

Findings, each earned by an iteration: the image-domain denoiser caps at ∼0.26; deeper *adjoint* data-consistency lowers hr (it re-injects streaks); the *filtered* gradient breaks the ceiling at 13 parameters; an in-loop prox (proximal / denoising step) hurts (the filtered gradient already reconstructs); a learned cross-step combination helps; a full conjugate-gradient DC solve does not beat

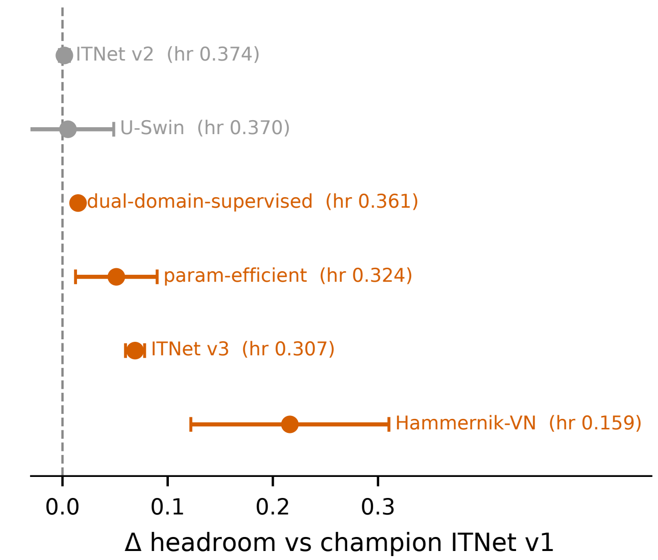

FIGURE S1 | Mayo top-solver significance: paired Δheadroom versus the champion (ITNet v1), mean ± 95 % CI over $n = 5$ patients. A confidence interval that crosses zero (grey) is a statistical tie with the champion; one that does not (red) is significantly worse. ITNet v2 and U-Swin tie; dual-domain-supervised, param-efficient, ITNet v3, and Hammernik-VN separate. Param-efficient's interval is wide — its per-patient deficit is noisy — which is why it ties on hr at the 1 % level despite a larger mean gap than dual-domain-supervised.

TABLE S5 | Parameter-efficient recombination arc on Mayo (low-dose, denoising). hr is validation headroom; the last row is the reported low-parameter pick.

step	architecture	params	val hr
CNN prox+DC (seed)	learnable CNN regulariser	~2800	0.24
FoE filter bank	field-of-experts reg (agent ceiling)	**1921**	**0.25** (ceiling)
+ Manduca projection filter	noise-adaptive sinogram bilateral	1926	0.394
+ multi-scale projection bank	parallel Manduca scales	1937	0.400 (best)
halved FoE (Pareto pick)	n_f 24 → 12	**969**	**0.360** (test 0.324)

TABLE S6 | Parameter-efficient recombination arc on breast (128-view sparse), all under the 20-min budget.

step	architecture	params	val hr
image-domain unroll (any config)	Mayo-style FoE denoiser	~1930	≤ 0.26 (ceiling)
filtered DC FBP($\mathbf{Ax} - \mathbf{g}$)	$K = 30$	**13**	**0.4345**
+ learned primal-dual combination	$W = 4$	183	0.5047
+ **3-scale bilateral output filter**	$\sigma_r = 0.02$	**195**	**0.6201** (single best)

the single filtered gradient under a fixed wall; and Wu-2015 band machinery adds nothing here. The champion is seed-fragile in the variational-network manner: 4 of 5 seeds cluster at 0.571 ± 0.033, one collapses; two stabilizers did not recover it. On the held-out test set the compact solver scores hr 0.6212±0.0076 — the tightest per-case std of any solver that clears the baseline (the only smaller stds belong to degenerate solvers whose hr is ≈ 0).

The gains are back-loaded, and human-cued — on both datasets. They are not spread evenly across the ~40-iteration searches. Under the agent alone the best value plateaued early — at hr ≈ 0.25 on Mayo (by iteration 7) and ≈ 0.50 on breast (by iteration 15) — and then sat flat for a dozen or more iterations. Only in the second half, after a human began suggesting untried directions, did each break through: Mayo to hr 0.40 from iteration 25,

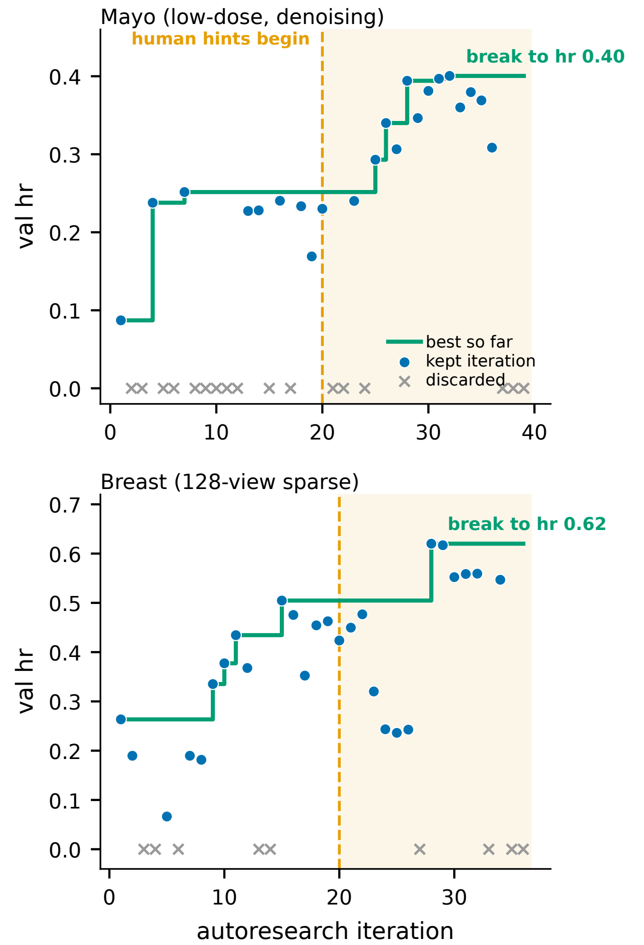

FIGURE S2 | Compact-solver search trajectories. Top: Mayo (low-dose). Bottom: breast (128-view sparse). Points are per-iteration validation headroom (discarded iterations shown at hr 0); the step line is the best value so far. On both datasets the agent's own first 20 iterations plateau (hr ≈ 0.25 Mayo, ≈ 0.50 breast); the decisive break comes only in the hinted second half (shaded) — to hr 0.40 and 0.62 respectively.

breast to hr 0.62 at iteration 28 (Figure S2). This is the clearest evidence in the study of the agent needing human steering to escape a local optimum; without the hints it would have reported both compact solvers at their plateaus.

Mayo vs breast contrast. On Mayo the compact optimum is an evolved field-of-experts plus multi-scale bilateral denoiser (hr 0.324 at ~0.00097 M params), statistically tied with the top tier at the 1 % level. On a denoising problem, a tiny denoiser suffices. On sparse-view breast, that same denoiser fails; data-consistency is the differentiator.[16] This problem-dependence is the practical face of the known-operator principle[33,34]: embedding the exact forward operator cannot raise, and usually lowers, the maximum error bound, and it curbs the hallucination risk of unconstrained learned maps.[35,36] The agent re-derived each optimum; it did not transfer.

Human role in this study. The recombination *strategy* was a human idea, on both datasets. On Mayo the human repeatedly asked whether more parameter-efficient approaches existed and explicitly suggested combining the pieces. On breast the finale was framed as a cross-method recombination and steered away from copying Mayo. The agent executed and mechanistically refined; it did not originate the recombination strategy.

S6 | Noise model, no-retrain, and retrained protocols

The full noiseless and noisy breast boards, the framework-axis reading of the reversal, and the retrained-board recovery are in the main text; this section records the exact protocols behind both noisy boards (no-retrain and retrained).

Noise model. We convert each clean line-integral sinogram $\mathbf{g}$ to expected photon counts $I_0\, e^{-\mathbf{g}}$, draw Poisson counts, and take the logarithm back to line integrals, following the standard low-dose forward model (Eq. (6) of the main text). The single parameter is the incident photon count I_0; we use $I_0 = 10^5$, a comparatively high dose, so the perturbation is deliberately mild — on the order of 1–2 % at the thickest (most attenuating) ray (the staging driver records the mean-absolute and maximum line-integral perturbation per split). The intent is not to stress-test at low dose but to ask whether a *small*, realistic departure from the idealised noiseless data already reorders the leaderboard. It does.

Determinism and fairness. The noise is drawn once per test case from a fixed seed and cached, so every method is evaluated on the byte-identical noisy input. Two solvers that change places do so because of how they process the same measurements, not because they happened to see different noise.

No-retrain board. Supervised solvers load the checkpoint they trained on the noiseless data and run inference only — their weights never see a noisy example. Classical and per-scene solvers (TV, PWLS-TV, bilateral, NAF, R^2-Gaussian) carry no pretrained weights and simply re-run their normal inference-time fit on the noisy input, which is their intended mode of use. The FBP baseline in the headroom metric is recomputed from the *noisy* sinogram, so hr measures each method's improvement over the noisy FBP and every method on the board is scored against the same baseline. The comparison is thus fair *within* the noisy board, even though the noisy and noiseless hr scales are not directly comparable.

Retrained board. For the retrained-noisy board (main text, "Retraining on the noise restores the ranking") we stage a deterministic Poisson-noised copy of the *train* and *validation* sinograms at the same $I_0 = 10^5$ and seed, then retrain each trainable solver's noiseless best-iteration configuration — unchanged architecture and hyper-parameters — from scratch on the noisy train split, select the checkpoint on the noisy validation split, and score it on the noisy test set. The truth stays clean throughout, so hr still measures recovery of the clean image; the FBP baseline is again recomputed from the noisy sinogram, so the retrained and no-retrain boards share the same scale and are directly comparable. Classical, per-scene, and zero-shot solvers have no trainable weights and are carried over unchanged from the no-retrain board. We retrained only the frozen clean-best configuration and did not re-search hyper-parameters for the noisy problem, so the retrained scores are a conservative lower bound on what matched-noise training could achieve. This isolates a single variable — clean versus noisy training data at a fixed configuration — so the recovery of the clean-favored supervised methods (e.g. dual-domain-supervised, hr $0.00 \rightarrow 0.958$) is attributable to the training distribution, not to any re-tuning.

S7 | Agent-behavior — extended account

This account follows the autoresearch paradigm of automated open-ended scientific discovery[37] realised through LLM code agents that resolve real software tasks[38,39] and physics-informed agentic development in adjacent imaging fields.[40] All iterations on both datasets were driven by a single language model, Claude Opus 4.8 (Anthropic; model `claude-opus-4-8`, 1M-token context); the model identifier is recorded in every iteration's provenance record. We summarise here what the agent did well, where it failed, and which parts of the study still required a human.

Strengths. The agent is fast from paper to code: it read a method description and produced a running solver in a fraction of a day (Section S2), which is what made 26 methods across two datasets tractable for one supervisor. It is a strong hyper-parameter optimiser, moving one knob at a time against the frozen metric and reliably driving each solver to its ceiling. It keeps a fixed compute budget without drift — the 20-minute per-iteration wall was honoured across every run, a discipline humans find tedious. It scales evaluation almost for free, launching and tracking dozens of independent benchmark jobs in parallel; this parallelism, not raw single-task speed, is its most valuable practical property. And it follows well-decomposed instructions faithfully, provided each sub-task is stated concretely.

Weaknesses. The agent does not invent new methods; it implements, tunes, and recombines existing ones, and it proposed no new reconstruction principle. It must be *forced* to recombine: left unsupervised it converges early and then pads — in one breast sub-run it replayed near-identical configurations until an audit caught the repetition. Its CT-image "vision" is unreliable: it repeatedly failed to see obvious streaks and artefacts in reconstructions and sinograms, so throughout the study the numeric metric, never the rendered image, had to be the source of truth. Long tasks must be decomposed into sub-tasks; without decomposition even a one-million-token context is exhausted before the task finishes. Finally, it overfits to the task and the chosen metric — though, as the reversal shows, that failing is not unique to agents; human researchers overfit to benchmarks too.

Human meta-strategies (repeatable, and the same on both datasets). Four human interventions recurred and mattered. *Persistence and breadth*: repeatedly asking the agent for more directions once it had stalled. *Recombination*: supplying the idea of assembling proven pieces into one compact solver — the decisive move behind the parameter-efficient result on both datasets (Section S5). *Redirection*: forbidding the agent from copying the other dataset's answer, which forced it to re-derive each optimum from local evidence. *Auditing*: catching padding, enforcing the compute budget, and repairing provenance. The defensible division of labour is therefore this: the agent is a tireless, mechanistically reasoning, self-modifying executor, and the human is the strategist and the auditor.

S8 | Additional results and data availability

Figure S3 reports the effect size (Cohen's d_z) against the raw Δhr to the champion on the breast board ($n = 200$), for the ten highest-ranked solvers. Effect size is only meaningful for solvers that clear the baseline, so we restrict it to ranked solvers with hr> 0. At $n = 200$ even small hr gaps are statistically significant, unlike the $n = 5$ Mayo board. The parameter/accuracy trade-off is Figure 2 of the

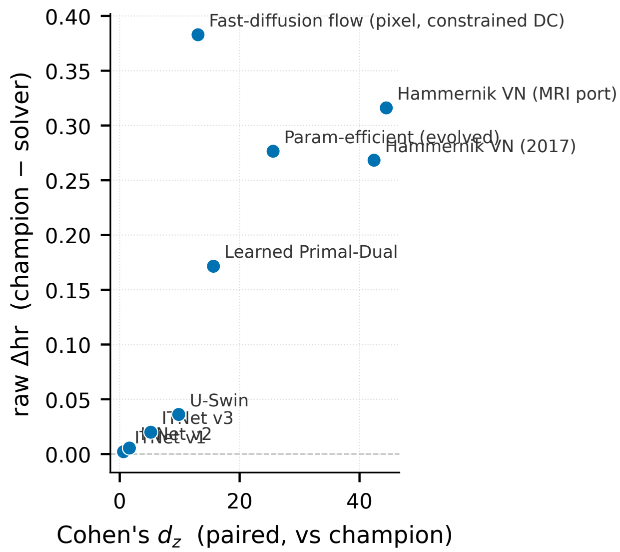

FIGURE S3 | Effect size on the breast board ($n = 200$): Cohen's d_z versus raw Δhr against the champion, for the ten highest-ranked solvers (nine non-champion points; all hr> 0). The breast board separates cleanly because of the sample size ($n = 200$, every pairwise $p < 10^{-4}$), not because the effects are large; at the $n = 5$ Mayo board the same-size gaps are not significant.

main text, and the per-step recombination climb is in Tables S5 and S6.

Data and code. The code, the 26 solvers and the compact recombination, and the per-iteration records are in the public repository, with the leaderboards on the project page:

https://github.com/akmaier/Agent4CT
https://akmaier.github.io/Agent4CT/

The dashboard lists the boards for all three datasets (Mayo, noiseless breast, noisy breast) with each solver's per-iteration trajectory. Every iteration stores its configuration, reconstruction, and metric, so each number in this paper can be traced to the run that produced it. The leaderboards and results in this paper correspond to the repository at commit `e52b55d8`. The repository also contains the Mayo solver×metric significance matrix, the noiseless-versus-noisy reconstruction montages, the per-solver design documents, and the `CONFIG` defaults.

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